



CS 540 Module Four Assignment Guidelines and Rubric

Overview

This assignment builds your ability to compare different types of visualizations that represent the same data set. The skill of selecting and defending a visualization is essential for professionals who need to present data clearly and effectively to a variety of audiences. Knowing how to choose the best visual format supports accurate interpretation and informed decision making.

In your field of study or workplace, you may be asked to justify why one chart is more effective than another when communicating results to stakeholders, clients, or team members. Practicing this kind of evaluation prepares you to make thoughtful choices that strengthen the impact and credibility of your data presentations.

Scenario

You've recently joined a data analytics team at a nonprofit organization that focuses on public policy. Your team is working on a report on regional education funding. Several team members have created different visualizations using the same data set, and leadership wants your input on which one should be featured in the report.

Your audience includes policymakers and education advocates, many of whom are not experts in data analysis. You must evaluate the visualizations and recommend which one communicates the data most effectively to this audience.

Directions

In this assignment, you will write a short analysis comparing multiple data visualizations that represent the same data set. Before you begin, review the data set and the three different visualizations located in the Module Four Assignment data folder in Codio. When you are ready to begin your work, access Codio by clicking the title of the next task.

Specifically, you must address the following rubric criteria:

1. Describe how each visualization **represents the data**.
2. Describe the **key characteristics** of each visualization. Include the following:
 - A. Chart type
 - B. Layout
 - C. Labeling
3. **Compare the visualizations** by analyzing how effectively each one communicates the data set's insights. Include the following:
 - A. Clarity
 - B. Data type alignment
4. Evaluate which visualization is most **appropriate** for the given audience. Include the following:

A. Rationale for your selection

5. Recommend multiple **refinements** to any of the visualizations provided. Include the following:

A. Justification of how the refinements would better suit the audience's needs

What to Submit

Submit your assignment as a 2- to 3-page Microsoft Word document with double spacing, 12-point Times New Roman font, and one-inch margins.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

Module Four Assignment Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Represent the Data	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes how each visualization represents the data	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Key Characteristics	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes the key characteristics of each visualization, including the chart type, layout, and labeling	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Compare the Visualizations	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Compares the visualizations by analyzing how effectively each one communicates the data set's insights, including clarity and data type alignment	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Appropriateness	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Evaluates which visualization is most appropriate for the given audience, including rationale for the selection	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Refinements	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Recommends multiple refinements to any of the visualizations provided, including justification of how the refinements would better suit the audience's needs	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	5
Total:					100%